

DATA ANALYTICS PROJECT • End-to-End Workflow

Customer Behavior & Purchase Pattern Analysis

A clean synthesis of raw retail transactional data processed in Python, structured using MySQL queries, and visualized through Power BI.

Database	Jupyter Notebook	BI Tool	Dataset Size
Project1	Project1.ipynb	Power BI Desktop	3,900 Records (~4K)

01 • CORE HIGHLIGHTS

TOTAL VOLUME

233K

3,900 Orders

AVG QUANTITY

59.76

Per Purchase

CUSTOMER BASE

4,000

Unique Users

SUBSCRIBERS

26.88%

63K Share

Key Finding: Clothing is the top performing category generating 104K in sales volume (3.40K customer interactions), followed by Accessories (74K) and Footwear (36K). Non-subscribers generate 73.12% (170K) of total volume, highlighting an opportunity for conversion.

02 • DATA PREPROCESSING (JUPYTER NOTEBOOK)

Raw data was sanitized and structured in `Project1.ipynb` prior to database ingestion:

Null & Rating Cleaning

Handling missing values across review ratings, shipping options, and discounts to prevent analytical bias.

Column Standardization

Converted column names to standard snake_case formats (purchase_amount, review_rating) for seamless SQL querying.

Feature Binning

Segmented customer ages into concise demographic groups (Young Adult, Adult, Middle Aged, Senior).

Type Formatting

Cast purchase volumes to numeric datatypes and binary flags to standardized string formats.

03 • DATABASE QUERIES & FINDINGS (`PROJECT1` DATABASE)

Queries executed on the `customer` table in MySQL Workbench:

Query 01 • Gender Revenue Share

What is the total revenue generated by male vs. female?

```
select gender, sum(purchase_amount)
from customer
group by gender;
```

Takeaway: Provides the overall volume distribution across gender segments to inform targeted marketing campaigns.

Query 02 • Discount Sensitivity

Which customer used a discount but still spent more than the average purchase amount?

```
select customer_id, purchase_amount
from customer
```

```
where discount_applied= 'Yes' and purchase_amount > (select avg(purchase_ammoun
```

Takeaway: Identifies high-value buyers who utilize discounts while maintaining basket sizes above the baseline average of 59.76.

Query 03 • Product Ratings

Which are the top 5 products with the highest average review rating?

```
select item_purchased, round(avg(review_rating),2) as "average_product_rating"
from customer
group by item_purchased
order by avg(review_rating) desc
limit 5;
```

Takeaway: Top-rated products include Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), and Skirt (3.78).

Query 04 • Shipping Comparison

Compare the average purchase amount between standard and express shipping.

```
select avg(purchase_amount), shipping_type
from customer
where shipping_type in ('Standard','Express')
group by shipping_type;
```

Takeaway: Compares basket sizes across fulfillment types to see if shipping speed correlates with larger order volumes.

Query 05 • Subscription Impact

Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.

```
select subscription_status,
count(customer_id) as total_customers,
avg(purchase_amount) as avg_spend,
sum(purchase_amount) as total_revenue
from customer
group by subscription_status
order by total_revenue, avg_spend desc;
```

Takeaway: Non-subscribers account for 170K of total volume compared to 63K from subscribers, proving subscription acquisition is a primary growth lever.

Query 06 • Discount Reliance

Which 5 products have the highest percentage of purchase with discount applied out of total purchase?

```
select item_purchased,  
100 * count(case when discount_applied = "Yes" then item_purchased end)/count(  
from customer  
group by item_purchased  
order by discount_purchase_percentage desc  
limit 5;
```



Takeaway: Highlights items that rely heavily on promotional discounts to move volume.

Query 07 • Customer Segmentation

Segment customers into new, returning and loyal based on their total number of previous purchases, and show the count of each segment.

```
with customer_type as(  
select customer_id, previous_purchases,  
case  
when previous_purchases=1 then "New"  
when previous_purchases>=2 and previous_purchases<=10 then "Returning"  
else "Loyal"  
end as customer_segment  
from customer  
)  
  
select customer_segment, count(*) as "Number of Customers"  
from customer_type  
group by customer_segment;
```

Takeaway: Categorizes the customer base into New, Returning, and Loyal cohorts for targeted retention logic.

Query 08 • Category Top Performers

What are the 3 most purchased items within each category?

```
with ranked_items as (  
select
```

```
        category,  
        item_purchased,  
        count(*) as total_purchases,  
        row_number() over(partition by category order by count(*) desc) as rank_  
from customer  
group by category, item_purchased  
)  
  
select  
    category,  
    item_purchased,  
    total_purchases  
from  
    ranked_items  
where  
    rank num <=3;
```

Takeaway: Uses partition window functions to uncover the top 3 best-selling products per category.



Query 09 • Repeat Buyer Subscription Propensity

Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

```
select  
    subscription_status,  
    count(customer_id) as repeated_buyers  
from customer  
where  
    previous_purchases > 5  
group by  
    subscription_status;
```

Takeaway: Measures subscription conversion rates among highly engaged repeat buyers (>5 orders).

Query 10 • Demographic Contribution

What is the revenue contribution of each age group?

```
select age_group, sum(purchase_amount) as total_revenue  
from customer  
group by age_group  
order by total_revenue desc;
```

Takeaway: Breaks down purchase volume across binned age brackets (Young Adult, Adult, Middle Aged, Senior).

04 • DASHBOARD INSIGHTS (POWER BI)

Sales By Category

Clothing leads with **104K**, followed by Accessories (**74K**), Footwear (**36K**), and Outerwear (**19K**).

Subscription Breakdown

Total volume of 233K split into **170K (73.12%)** non-subscribers and **63K (26.88%)** subscribers.

Customer Volume by Category

Clothing interactions hit **3.40M**, Accessories **2.41M**, Footwear **1.17M**, Outerwear **0.62M**.

Review Ratings

Consistently high quality across products, led by Gloves (3.86), Sandals (3.84), Boots (3.82), and Hat (3.80).

05 • RECOMMENDATIONS

- 1

Subscription Conversion Push: Convert the 73.12% non-subscriber majority by targeting repeat buyers (>5 orders) with exclusive subscriber perks.
- 2

Double Down on Top Categories: Focus inventory and hero placement on Clothing (104K volume) and Accessories (74K volume).
- 3

Minimum Spend Thresholds: For customers spending above average (59.76) with discounts, shift to threshold promotions (e.g., "10 off on orders over 75") to raise order values.

